

1. Largest of Three Numbers

Buggy Code:

```
int a=10,b=20,c=30;
if(a>b || a>c) cout<<a;
else if(b>a || b>c) cout<<b;
else cout<<c;
```

Correct Code:

```
int a=10,b=20,c=30;
if(a>b && a>c) cout<<a;
else if(b>a && b>c) cout<<b;
else cout<<c;
```

2. Leap Year

Buggy Code:

```
int year=1900;
if(year%4==0) cout<<"Leap Year";
else cout<<"Not Leap Year";
```

Correct Code:

```
int year=1900;
if((year%4==0 && year%100!=0) || year%400==0)
    cout<<"Leap Year";
else
    cout<<"Not Leap Year";
```

3. Even Numbers (1 to 50)

Buggy Code:

```
for(int i=1;i<=50;i++)
if(i%2==1) cout<<i<<" ";
```

Correct Code:

```
for(int i=1;i<=50;i++)
if(i%2==0) cout<<i<<" ";
```

4. Factorial

Buggy Code:

```
int num=5,fact=1;
for(int i=0;i<=num;i++)
fact*=i;
cout<<fact;
```

Correct Code:

```
int num=5,fact=1;
for(int i=1;i<=num;i++)
fact*=i;
cout<<fact;
```

5. Prime Number

Buggy Code:

```
int num=9;
for(int i=2;i<num;i++)
if(num%i!=0){ cout<<"Prime"; break;}
```

Correct Code:

```
int num=9; bool prime=true;
if(num<=1) prime=false;
for(int i=2;i<=num/2;i++)
if(num%i==0){ prime=false; break;}
cout<<(prime?"Prime":"Not Prime");
```

6. Count Digits

Buggy Code:

```
int num=0,count=0;
while(num>0){ count++; num/=10;}
cout<<count;
```

Correct Code:

```
int num=0,count;
if(num==0) count=1;
else{ count=0; while(num>0){count++; num/=10;}}
cout<<count;
```

7. Sum of Digits

Buggy Code:

```
int num=456,sum;
while(num>0){ sum+=num%10; num/=10;}
cout<<sum;
```

Correct Code:

```
int num=456,sum=0;
while(num>0){ sum+=num%10; num/=10;}
cout<<sum;
```

8. Reverse Number

Buggy Code:

```
int num=123,rev;
while(num>0){ rev=rev*10+num%10; num/=10;}
cout<<rev;
```

Correct Code:

```
int num=123,rev=0;
while(num>0){ rev=rev*10+num%10; num/=10;}
cout<<rev;
```

9. Palindrome Number

Buggy Code:

```
int num=121, rev=0;
while(num>0){ rev=rev*10+num%10; num/=10;}
if(num==rev) cout<<"Palindrome";
else cout<<"Not Palindrome";
```

Correct Code:

```
int num=121, temp=num, rev=0;
while(temp>0){ rev=rev*10+temp%10; temp/=10;}
if(num==rev) cout<<"Palindrome";
else cout<<"Not Palindrome";
```

10. Fibonacci Series

Buggy Code:

```
int a=0, b=1, c;
for(int i=1; i<=5; i++){
c=a+b; cout<<c<<" ";
a=b+c; b=c;}
```

Correct Code:

```
int a=0, b=1, c;
for(int i=1; i<=5; i++){
cout<<a<<" ";
c=a+b; a=b; b=c;}
```